

JEE MAIN REVISION MASTER: CARBOXYLIC ACIDS AND ITS DERIVATIVES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Oxidation, Reduction & Identification Tests	6 PYQs	Tests & Reductions
Moderate	Nucleophilic Addition & Carbonyl Condensation	5 PYQs	Carbonyl Reactions

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Oxidation, Reduction & Identification Tests	$\text{Iodoform test is positive for } \text{CH}_3\text{CO-} \text{ or } \text{CH}_3\text{CH(OH)- groups} \implies \text{CHI}_3 \text{ (yellow ppt)}$
Nucleophilic Addition & Carbonyl Condensation	$\text{Aldol: Requires } \alpha\text{-H} \to \alpha,\beta\text{-unsaturated carbonyl}$, $\text{Cannizzaro: No } \alpha\text{-H} \to \text{Alcohol} + \text{Carboxylate salt}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Oxidation, Reduction & Identification Tests (Moderate Yield — 6 PYQs)

Core Principle: Silver mirror Tollens' test, Fehling's cuprous reduction, yellow iodoform precipitation, Clemmensen (Zn-Hg/HCl), and Wolff-Kishner (NH₂NH₂/KOH) carbonyl deoxygenation.

■ **Caution / JEE Pitfall:** Benzaldehyde reduces Tollens' reagent to give silver mirror, but does NOT reduce Fehling's solution!

■ **Speed Trick:** Clemmensen reduction is suitable for acid-stable compounds. For acid-sensitive carbonyls, use Wolff-Kishner reduction (basic medium).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Carboxylic acids are more acidic than phenols because..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Statement - I : Reduction of ester by DIABL-H followed by hydrolysis gives aldehyde. Statement - II : Oxidation of benzyl alcohol with aqueo..." [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "Ethanoic acid undergoes Hell-Volhard Zelinsky reaction but methanoic acid does not, because of..." [Ans: N/A]

3.2 Nucleophilic Addition & Carbonyl Condensation (Moderate Yield — 5 PYQs)

Core Principle: Reversible nucleophilic additions, Aldol condensation mechanism, and Cannizzaro disproportionation of non-enolizable aldehydes.

■ **Caution / JEE Pitfall:** Benzaldehyde and Formaldehyde lack α -hydrogen atoms, so they undergo CANNIZZARO reaction with concentrated NaOH, NOT Aldol condensation!

■ **Speed Trick:** Reactivity toward nucleophilic addition: $\text{HCHO} > \text{CH}_3\text{CHO} > \text{CH}_3\text{COCH}_3$ (decreases with steric crowding and +I electron donation).

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "Which of the following has most acidic Hydrogen ?..." [Ans: N/A]
- [2022 Shift] Q#4 (MCQ): "Reaction by which benzaldehyde cannot be prepared is..." [Ans: N/A]
- [2021 Shift] Q#4 (MCQ): "The product 'A' gives white precipitate when treated with bromine water. The product 'B' is treated with barium hydroxide to give th..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Carboxylic acids are more acidic than phenols because

- (A) Formation of dimers
- (B) Intermolecular hydrogen bonding
- (C) More covalent nature
- (D) More resonance stabilisation of their conjugate base

Correct Answer: None

Solution Analysis: Carboxylic acids are more acidic than phenols because their conjugate base is more resonance stabilised. So the correct option is: Option D: More resonance stabilisation of their conjugate base When a carboxylic acid loses a proton, it forms a carboxylate ion: $\text{RCOOH} \rightarrow \text{RCOO}^- + \text{H}^+$ In the carboxylate ion, the negative charge is delocalised equally over two oxygen atoms: $\text{R}-\text{C}(=\text{O})-\text{O}^- \leftrightarrow \text{R}-\text{C}(\text{O}^-)=\text{O}$ Both resonance structures are equivalent, so the carboxylate ion is highly stable. Phenol also forms a conjugate base, called phenoxide ion:

Benchmark Problem 2: KCET 2025 (Q#3)

Which of the following has most acidic Hydrogen ?

- (A) Propanoic acid
- (B) Dichloroacetic acid
- (C) Trichloroacetic acid
- (D) Chloroacetic acid

Correct Answer: None

Solution Analysis: Trichloroacetic acid (Option C) is the strongest acid. Here's why: Electron-withdrawing effect Chlorine atoms pull electron density away from the $-\text{COOH}$ group, stabilizing the conjugate base (the carboxylate anion). More Cl $\rightarrow$ greater stabilization $\rightarrow$ stronger acid. Comparative pKa values Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$: pKa $\approx$ 4.87 Chloroacetic acid, ClCH_2COOH : pKa $\approx$ 2.86 Dichloroacetic acid, Cl_2CHCOOH : pKa $\approx$ 1.25 Trichloroacetic acid, CCl_3COOH : pKa $\approx$ 0.66 Since trichloroacetic acid has the lowest pKa, it donates its proton most readily and is thus the most acidic.